

开源软件基础

利用 Stack Overflow

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Outline

- 1 Stack Overflow 简介
- 2 How do I ask a good question
- 3 Stack Overflow Survey
- 4 Something Interesting

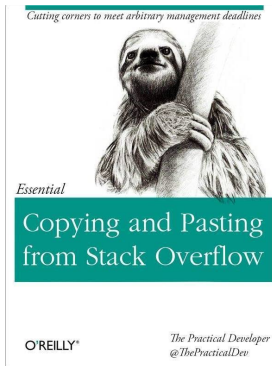
Stack Overflow 简介

Stack Overflow

Stack Overflow 是一个程序设计领域的问答网站，隶属 Stack Exchange Network。网站允许注册用户提出或回答问题，利用积分制度让用户自行管理。另外网站也会根据用户的贡献颁发徽章。用户创建的内容都使用知识共享协议授权。直至 2018 年 9 月，Stack Overflow 有超过 9,400,000 名注册用户和超过 16,000,000 个问题，其中最常见的主题有 JavaScript、Java、C#、PHP、Android、Python、jQuery 和 HTML^a

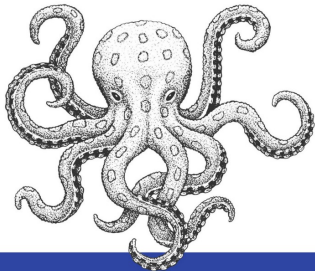
^ahttps://zh.wikipedia.org/zh-cn/Stack_Overflow

Stack Overflow



Stack Overflow

Just memorize these fourteen contextually dependant instructions



Exiting Vim

Eventually

ORLY?

@ThePracticalDev

How do I exit the Vim editor¹

¹<https://stackoverflow.com/questions/11828270/how-do-i-exit-the-vim-editor>

Stack Overflow

The internet will make those bad words go away



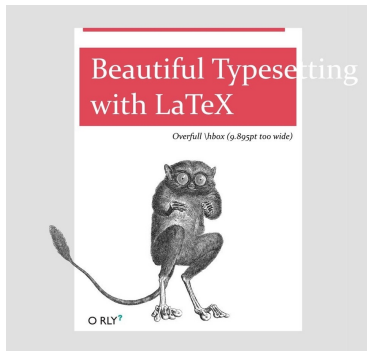
Essential

Googling the Error Message

ONLY?

The Practical Developer
@ThePracticalDev

Stack Overflow



Stack Overflow 简介

技术

Stack Overflow 是以 ASP.NET 3.5 撰写而成并使用 ASP.NET MVC (MVC 模式) 框架。未注册用户可以访问网站的大部分功能，使用 OpenID 登录的用户可以使用更多功能，诸如：创建个人文件、赚取声望点数并激活修改标签或者投票关闭问题。

Stack Overflow

声望值和管理制度

利用“声望值”来决定用户的管理权限，以此达到社区自我管理。声望值（Reputation）就是用户进行网站交互时能获取的分数，例如，用户 A 回答了一个问题，用户 B 对用户 A 的解答给予了“加分”，用户 A 就会因而获得 10 点声望值。对已有问题或答案扣分，作者和评价者的声望值都会降低，以此避免恶意报复。当声望值达到某个程度，用户的权限（Privileges）就会增加，如声望值超过 50 点就可以评论答案、声望值超过 2000 可以直接编辑问题，无需作者同意。虽然“声望值”很大程度能让社区高效的自我管理，但是声望值机制并非完美。例如用户询问或回答新手常遇到的简单问题（在有大量的用户的语言），就很容易获得非常多的声望值，导致他们获得不符合自身能力的 Privileges。

用户可以提交编辑标题、提交编辑主题标签、提交编辑问题内文（如果声望值不够高，就需要发问者接受编辑）。并且能够将问题的状态标示为重复问题、低质量问题、删除（标示为被删除）问题、关闭问题，让社区的问答维持质量。在 2016 年，社区删除了 150 万个帖子，其中大约 8% 是由版主删除的。

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How to ask question²

- 1 Write a title that summarizes the specific problem
- 2 Introduce the problem before you post any code
- 3 Help others reproduce the problem
- 4 Include all relevant tags
- 5 Proof-read before posting!
- 6 Post the question and respond to feedback

²<https://stackoverflow.com/help/how-to-ask>

What to avoid asking

To prevent your question from being flagged and possibly removed, avoid asking subjective questions where ...

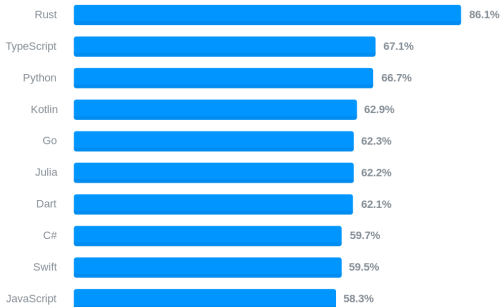
- ① every answer is equally valid: “What’s your favorite _____?”
- ② your answer is provided along with the question, and you expect more answers: “I use _____ for _____, what do you use?”
- ③ there is no actual problem to be solved: “I’m curious if other people feel like I do.”
- ④ you are asking an open-ended, hypothetical question: “What if _____ happened?”
- ⑤ your question is just a rant in disguise: “_____ sucks, am I right?”

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The Beloved

Rust held onto it's spot as the most beloved language among the professional developers we surveyed. That said, the majority of developers who took the survey aren't familiar with the language³.



³<https://stackoverflow.blog/2020/05/27/2020-stack-overflow-developer-survey-results/>

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StackSort

INEFFECTIVE SORTS

```

DEFINE HALFHEARTEDMERGESORT(LIST):
  IF LENGTH(LIST) < 2:
    RETURN LIST
  PIVOT = INT(LENGTH(LIST) / 2)
  A = HALFHEARTEDMERGESORT(LIST[:PIVOT])
  B = HALFHEARTEDMERGESORT(LIST[PIVOT:])
  // UMMMMMM
  RETURN [A, B] // HERE. SORRY.

```

```

DEFINE FASTBOGOSORT(LIST):
  // AN OPTIMIZED BOGOSORT
  // RUNS IN O(N LOG N)
  FOR N FROM 1 TO LOG(LENGTH(LIST)):
    SHUFFLE(LIST):
    IF ISORTED(LIST):
      RETURN LIST
  RETURN "KERNEL PAGE FAULT" (ERROR CODE: 2)

```

```

DEFINE JOBINERNEWQUICKSORT(LIST):
  OK SO YOU CHOOSE A PIVOT
  THEN DIVIDE THE LIST IN HALF
  FOR EACH HALF:
    CHECK TO SEE IF IT'S SORTED
    NO WAIT, IT DOESN'T MATTER
    COMPARE EACH ELEMENT TO THE PIVOT
    THE BIGGER ONES GO IN A NEW LIST
    THE EQUAL ONES GO INTO, UH,
    THE SECOND LIST FROM BEFORE
  HANG ON, LET ME NAME THE LISTS
  THIS IS LIST A
  THE NEW ONE IS LIST B
  PUT THE BIG ONES INTO LIST B
  NOW TAKE THE SECOND LIST
  CALL IT LIST, UH, A2
  WHICH ONE WAS THE PIVOT IN?
  SCRATCH ALL THAT
  IT JUST RECURSIVELY CALLS ITSELF
  UNTIL BOTH LISTS ARE EMPTY
  RIGHT?
  NOT EMPTY, BUT YOU KNOW WHAT I MEAN
  AM I ALLOWED TO USE THE STANDARD LIBRARIES?

```

```

DEFINE PANICSORT(LIST):
  IF ISORTED(LIST):
    RETURN LIST
  FOR N FROM 1 TO 10000:
    PIVOT = RANDOM(0, LENGTH(LIST))
    LIST = LIST[:PIVOT] + LIST[PIVOT:]
    IF ISORTED(LIST):
      RETURN LIST
  IF ISORTED(LIST):
    RETURN LIST
  IF ISORTED(LIST): // THIS CAN'T BE HAPPENING
    RETURN LIST
  IF ISORTED(LIST): // COME ON COME ON
    RETURN LIST
  // OH JEEZ
  // I'M GONNA BE IN SO MUCH TROUBLE
  LIST = [ ]
  SYSTEM("SHUTDOWN -h +5")
  SYSTEM("RM -rf /")
  SYSTEM("RM -rf /*")
  SYSTEM("RM -rf /")
  SYSTEM("dd if=/dev/urandom of=/dev/null") // PORTABILITY
  RETURN [1, 2, 3, 4, 5]

```

StackSort connects to StackOverflow, searches for "sort a list", and downloads and runs code snippets until the list is sorted⁴. JS⁵/Python implementation⁶

⁴<https://xkcd.com/1185/>

⁵<http://gkoberger.github.io/stacksort/>

⁶<https://pypi.org/project/stacksort/>

StackSort

```
>>> from stacksort import quicksort
>>> quicksort([1, 3, 2, 4])
[1, 2, 3, 4]
>>> from stacksort import heap_sort
>>> heap_sort([1, 2, 4, 3])
[2, 3, 4]
```